



M-Tech Production & Marketing

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Experiment 1

LAB TASKS

Internship Program — Batch 2026

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1. Experiment 1 – Lab Tasks

1.1 Viva Questions

1. Explain the relationship between Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL).

Artificial Intelligence (AI) is the broad field of computer science focused on developing systems capable of performing tasks that typically require human intelligence, such as reasoning, problem-solving, and decision-making. Machine Learning (ML) is a subset of AI that enables computers to learn patterns from data and improve their performance without being explicitly programmed. Deep Learning (DL) is a specialized branch of ML that utilizes multi-layered artificial neural networks to solve complex problems, including image recognition, speech processing, and natural language understanding.

2. Why is Anaconda commonly used in Data Science and Machine Learning?

Anaconda is a comprehensive Python distribution designed specifically for Data Science, Machine Learning, and Artificial Intelligence development. It comes with Python, numerous pre-installed scientific libraries, and package management tools. Additionally, it includes Jupyter Notebook and Anaconda Navigator, providing an efficient environment for coding, data analysis, visualization, and model development.

3. What is a Jupyter Notebook cell, and how is it executed?

A Jupyter Notebook is composed of individual cells that can contain either executable Python code or formatted text using Markdown. Code cells are executed by pressing **Shift + Enter**, allowing users to run code interactively and view the output immediately below the cell.

4. What does the `input()` function return, and why are `int()` and `float()` used?

The `input()` function accepts user input and returns the entered value as a **string** by default. When numerical calculations are required, the input must be converted into the appropriate numeric data type using functions such as `int()` for integers or `float()` for decimal numbers.

5. State the purpose of the following Python libraries.

- **NumPy:** Used for high-performance numerical computing, mathematical operations, and efficient array manipulation.
- **Pandas:** Used for data manipulation, cleaning, analysis, and working with structured datasets such as tables and spreadsheets.
- **Scikit-learn:** A powerful Machine Learning library used for data preprocessing, model training, prediction, evaluation, and implementing various ML algorithms.

6. Differentiate between Jupyter Notebook and Google Colab.

Jupyter Notebook	Google Colab
Runs locally on the user's computer.	Runs entirely in the cloud through a web browser.
Requires Python and libraries to be installed.	No installation is required.
Uses the computer's local resources.	Provides free access to cloud-based CPUs, GPUs, and TPUs.
Suitable for offline development.	Ideal for collaborative and cloud-based Machine Learning projects.

2. Multiple Choice Questions (MCQs)

Q1. Machine Learning is a subset of:

Answer: (B) Artificial Intelligence

Q2. Which tool installs Python along with Data Science libraries in one package?

Answer: (B) Anaconda

Q3. Which keyboard shortcut is used to execute a Jupyter Notebook cell?

Answer: (B) Shift + Enter

Q4. The `input()` function returns a value of which data type?

Answer: (C) String

Q5. Which Python library is primarily used for numerical array operations?

Answer: (B) NumPy

3. Intern Information

Field	Details
Intern Name	Abdullah Rafique
Supervisor Name	Mr. Mohazin Zahid
Date	06 July, 2026
Domain	AI/ML Intern
Intern Signature	<u>Abdy</u>

```
1  # Take input from the user
2  num1 = float(input("Enter first number: "))
3  num2 = float(input("Enter second number: "))
4
5  # Perform calculations
6  print("Sum =", num1 + num2)
7  print("Difference =", num1 - num2)
8  print("Product =", num1 * num2)
9  print("Quotient =", num1 / num2)
```

```
L1  # Take temperature in Celsius
L2  celsius = float(input("Enter temperature in Celsius: "))
L3
L4  # Convert to Fahrenheit
L5  fahrenheit = (celsius * 9 / 5) + 32
L6
L7  # Display result
L8  print("Temperature in Fahrenheit =", fahrenheit)
```

```
20  # Take user's name
21  name = input("Enter your name: ")
22
23  # Print greeting five times
24  for i in range(5):
25      print("Hello", name)
```